

User Query and Available Tool Schemas

“Kiểm tra phạt nguội cho xe máy biển số 59P1-12345 tại TP. Hồ Chí Minh.”

EN: “Check traffic fines for license plate 59P1-12345”

Method 1: End-to-End SLM

Tool schemas in prompt

Qwen3.5-2B/4B + QLoRA SFT

Autoregressive decoding

Custom `<tool_call>` tags

Rule-based parser

Structured tool call

Method 2: Bi-Encoder-Cross-Encoder

BGE-M3: semantic tool retrieval

Abstention gate and Top-k selection ($k \leq 3$)

XLNet: schema-aware parameter extraction

has_value

span

enum

boolean

Value normalization and deterministic assembly

Structured tool call